

Online Appendix

Accounting for GPU-Collateralized Debt: Depreciation, Revenue Recognition, and Off-Balance-Sheet Measurement

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Abstract

This online appendix accompanies Gibson, Hall, and Demerjian (2026). It provides materials omitted from the main paper for length: (i) GPU-collateralized debt market chronology and accumulation figure; (ii) circular-flow network map of counterparties; (iii) extended facility-level detail; (iv) warrant classification and fair-value technical notes; and (v) fire-sale recovery benchmarks and reduced-form pricing of the short-life state. Main-paper Appendix A retains ROIC, WACC, and uncommenced-lease present-value assumptions.

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1 Purpose and relationship to the main paper

The main paper documents three measurement channels (ASC 360 depreciation, ASC 606/718 customer warrants, ASC 842 uncommenced leases) using CoreWeave’s FY2025 10-K. For *Accounting Horizons* length,

extended institutional and technical material is placed here. Appendix A of the main paper (ROIC, WACC, lease PV) is **not** duplicated.

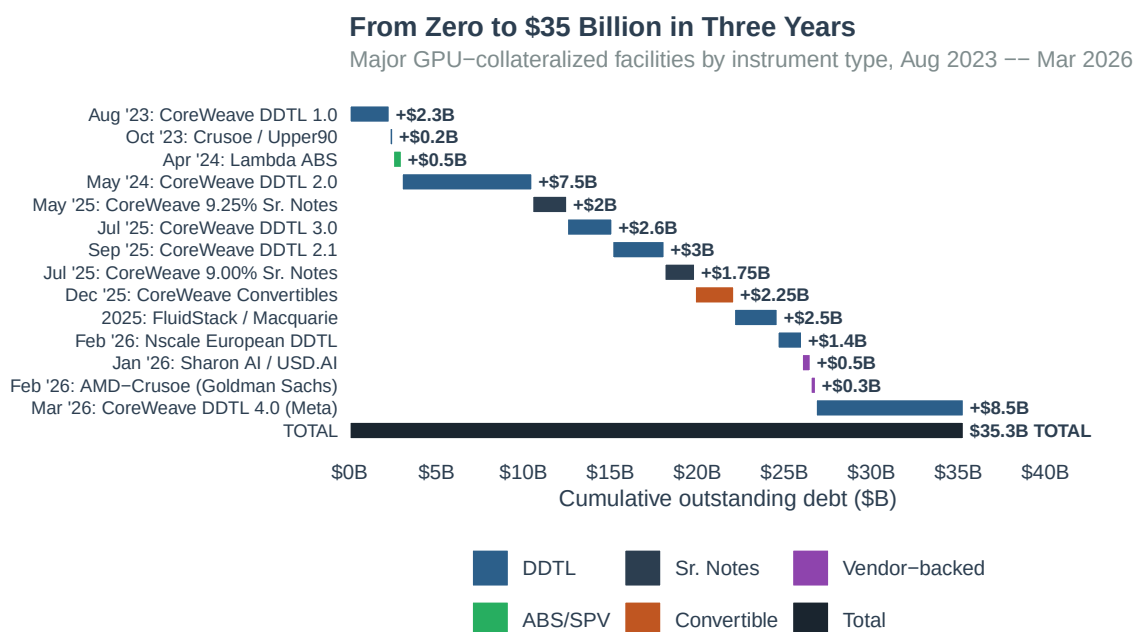
OA section	Main-paper pointer
OA.1 Market chronology	§2.3
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Supporting raw materials (schema, network build scripts, fire-sale framework notes) are in the `data/` folder of this repository.

2 Market chronology and accumulation

GPU-collateralized lending filled a financing gap: AI infrastructure firms needed billions for NVIDIA GPUs without conventional bank access. The watershed was CoreWeave’s August 2023 \$2.3 billion delayed-draw term loan (DDTL) led by Magnetar Capital and Blackstone Tactical Opportunities, with participation from BlackRock, PIMCO, Carlyle, Coatue, and DigitalBridge Credit (CoreWeave, Inc. 2023). Secured by NVIDIA GPUs and long-term customer contracts, that facility set the sector template.

Scaling was rapid. Crusoe borrowed \$200 million in October 2023 against approximately 20,000 GPUs. Lambda secured a \$500 million GPU financing SPV in April 2024. CoreWeave closed DDTL 2.0 at \$7.5 billion in May 2024. By late 2024, sector GPU debt exceeded \$11 billion (Financial Times 2024). Through 2025–early 2026: FluidStack / Macquarie capacity above \$10 billion; Nscale \$1.4 billion European DDTL; Crusoe \$300 million AMD-guaranteed loan (first major non-NVIDIA facility); Firmus \$10 billion authorized for Australian AI infrastructure (Bloomberg News 2026b). In March 2026 CoreWeave pursued Meta-backed DDTL 4.0 at approximately SOFR+225 bps, supported by more than \$19 billion in Meta contracts (Bloomberg News 2026a).



Source: Company filings, Financial Times, Bloomberg, author compilation

Figure 1: GPU-collateralized debt accumulation by instrument type, Aug 2023–Mar 2026. Amounts are drawn or committed (not authorized limits). DDTL 4.0 (Meta-backed) pending as of March 2026.

Instrument taxonomy. DDTLs pair GPU collateral with contracted revenue. Vendor-backed guarantees (NVIDIA capacity backstops; AMD’s Crusoe guarantee) transfer collateral risk back to the product-cycle originator. Senior notes are fixed-rate refinancing; convertibles trade coupon for dilution; ABS / tokenized structures push chip credit to investors furthest from technology risk. Each structure allocates useful-life,

demand, and refinancing risk to a different balance sheet—which is why measurement of those exposures matters for reporting and credit analysis.

3 Circular-flow network of counterparties

The contract chain is not linear. The same counterparties can appear as suppliers, customers, equity investors, guarantors, and lenders. A simplified chain:

1. **NVIDIA** sells GPUs to CoreWeave under take-or-pay arrangements.
2. **CoreWeave** pledges GPUs as collateral through ring-fenced SPVs.
3. **SPVs** secure private-credit debt.
4. **Lenders** (Blackstone, PIMCO, Magnetar, and others) fund facilities backstopped by hyperscaler contracts.
5. **Hyperscalers** (Microsoft, Meta, Oracle, and others) guarantee or anchor revenue via take-or-pay compute contracts.
6. Customer revenue services the debt; collateral values depend on NVIDIA's product cycle; NVIDIA revenue depends on continued neocloud purchases.

The circularity creates correlated default risk: standard models that treat defaults as independent are misspecified for this structure. Historical analogy: Carpenter and Lazonick (2023) documents Cisco's vendor-financing boom and subsequent write-offs when demand failed—the vendor financed its own demand.

Circular contract chain in GPU lending

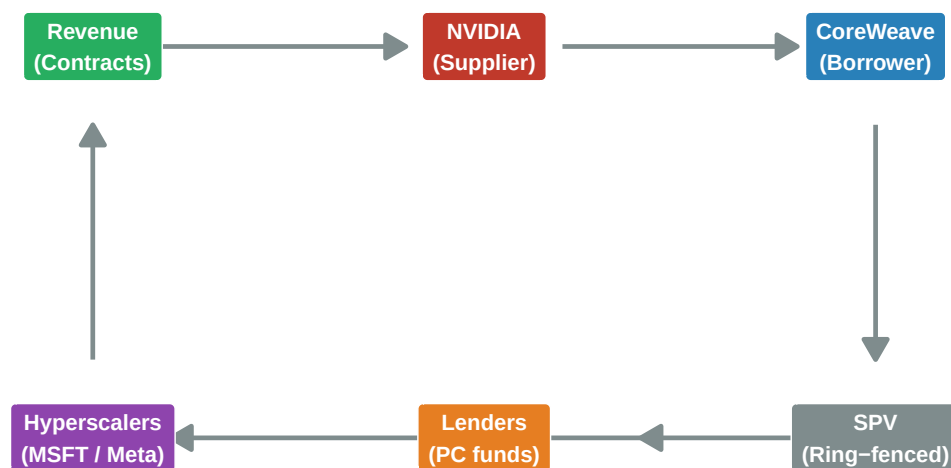


Figure 2: Simplified circular contract chain in GPU lending. Full directed network graph from the curated relationship database is shown next.

GPU Circular Capital Flows

Directed graph — edge width ~ deal size (log scale)

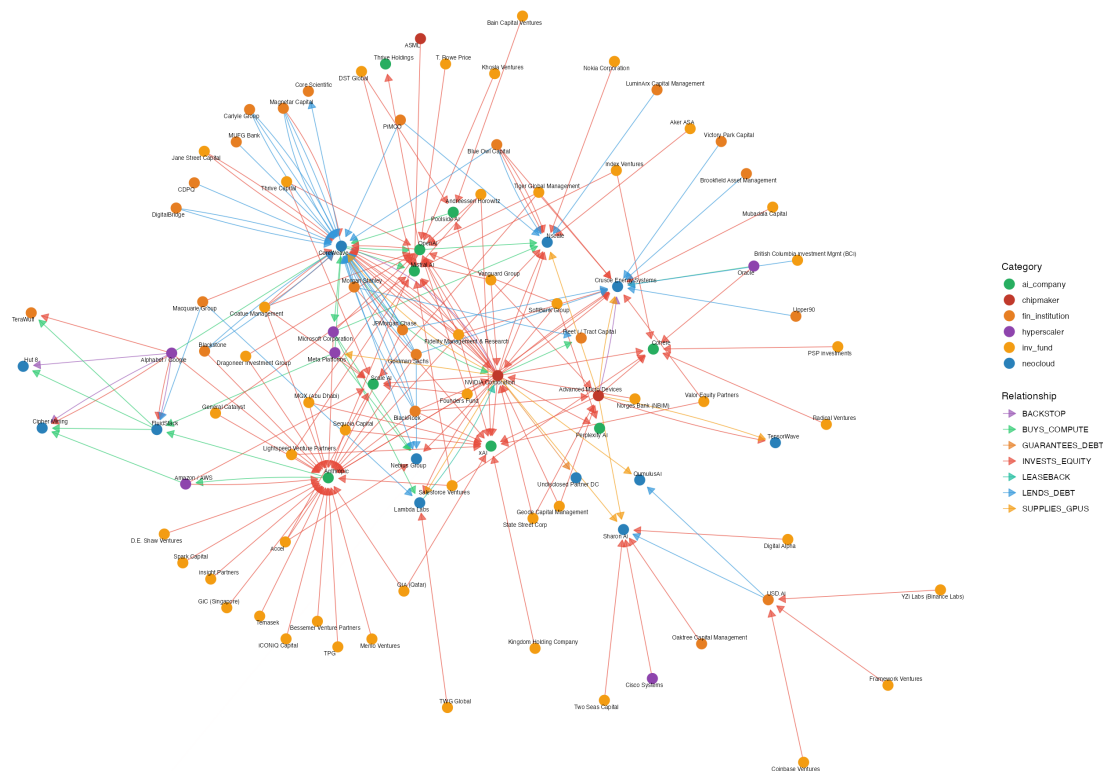


Figure 3: Full circular-flow network constructed from documented financing, supply, equity, and capacity-backstop relationships (author-curated database). Build scripts and schema: [online-appendix/data/](#).

Bankruptcy remoteness. CoreWeave DDTLs sit in wholly owned SPVs (e.g., CoreWeave Compute Acquisition Co. entities) with perfected first-priority security interests in GPUs, contracts, and SPV equity (10-K Note 10). Whether that structure survives Chapter 11—automatic stay under 11 U.S.C. §362; substantive consolidation—is untested. Take-or-pay enforceability is a separate vulnerability: termination provisions for material breach, delivery failure, and financing contingencies mean \$60.7 billion of remaining performance obligations is contractual face value, not risk-adjusted cash flow.

Repository files: `data/circular_flow_schema.sql`, `data/build_circular_flow_db.py`, `data/circular_flow_graph.R`, `data/circular_flow_research.md`.

4 Extended facility-level detail

Main-paper Table 1 summarizes CoreWeave’s twelve instruments (\$21.6 billion drawn). The table below expands to sector facilities with dates, maturities, rates, and sources where available.

Table 1: Extended GPU-collateralized facility data. Amounts are drawn or committed unless noted. CoreWeave rows (highlighted in source data) total \$21.6B drawn principal in the main paper.

Borrower	Facility	Outstanding	Date	Maturity	Rate	Source
CoreWeave	DDTL 1.0	\$1,553M	Jul 2023	Mar 2028	15%	10-K
CoreWeave	DDTL 2.0	\$5,037M	May 2024	Aug 2030	10%	10-K
CoreWeave	DDTL 2.1	\$2,741M	Sep 2025	Dec 2030	9%	10-K
CoreWeave	DDTL 3.0	\$340M	Jul 2025	Aug 2030	9%	10-K
CoreWeave	2030 Sr. Notes	\$2,000M	May 2025	Jun 2030	9.25%	10-K
CoreWeave	2031 Sr. Notes	\$1,750M	Jul 2025	Feb 2031	9.00%	10-K
CoreWeave	2031 Convertible	\$2,588M	Dec 2025	Dec 2031	1.75%	10-K
CoreWeave	Conv. Promissory	\$168M	Oct 2025	Apr 2026	7%	10-K
CoreWeave	RCF	\$1,000M	Pre-IPO	Nov 2029	6%	10-K
CoreWeave	OEM/Software	\$4,165M	Various	Mar 26–Jul 30	10%	10-K
CoreWeave	Magnetar Loan	\$273M	2025	Jan 2029	12%	10-K
CoreWeave	2024 Term Loan	Repaid	2024	Apr 2025	12%	10-K
Applied Digital	Sr. Secured 9.25%	\$2,350M	2024	NR	9.25%	APLD filings
Applied Digital	Convertible 2.75%	\$375M	Oct 2024	NR	2.75%	APLD filings
Lambda Labs	ABS/SPV	\$500M	Apr 2024	NR	NR	Press
Crusoe Energy	Term Loan	\$200M	Oct 2023	NR	NR	Press
Crusoe Energy	AMD-backed	\$300M	Feb 2026	NR	6%	Bloomberg
FluidStack	DDTL (Macquarie)	\$2.5B drawn	2025	NR	NR	Press
Nscale	European DDTL	\$1,400M	Feb 2026	NR	NR	Press
Sharon AI	Tokenized (USD.AI)	\$500M auth	Jan 2026	NR	NR	Press
Firmus	Infrastructure DDTL	\$10B auth	Feb 2026	NR	NR	Press
CoreWeave	DDTL 4.0 (Meta)	\$8,500M (pending)	Mar 2026	Pending / not disclosed	SOFR+225	Bloomberg

5 Warrant classification and fair-value technical notes

Main-paper §5.2 reports measurement outcomes for AMD’s Meta and OpenAI warrants under ASC 606-10-32-25 and ASC 718 as clarified by ASU 2025-04. This section records classification and valuation notes omitted for length.

5.1 Liability versus equity classification

Under ASC 718-10-35-2, share-based payments classified as **liabilities** are remeasured at fair value each period; **equity** awards are measured at grant date. AMD’s warrants have a fixed share count (320 million) and fixed exercise price (\$0.01), consistent with equity classification under ASC 480 and ASC 815-40 if freestanding (instrument indexed to AMD’s own stock). If equity-classified, the revenue reduction is fixed at measurement date; if liability-classified (e.g., because milestone vesting makes the number of shares variable in a way that fails equity classification), the reduction marks to market each period. AMD has not disclosed classification or net-of-warrant revenue.

5.2 Intrinsic value versus option value

The main paper’s \$61.8 billion figure is **intrinsic value** at approximately \$193 per share (March 2026): $(193 - 0.01) \times 320$ million. Under ASC 718-10-30-2, nonemployee awards are measured at fair value, which for warrants generally requires an option model. For deep-in-the-money penny warrants, Black–Scholes converges to intrinsic value: with roughly 50% equity volatility, five-year term, and 4.25% risk-free rate, model value per warrant is approximately \$192.60 versus intrinsic \$192.99 (difference < 0.3%). Intrinsic value is therefore an appropriate approximation here; higher strikes or shorter terms would reopen the gap.

5.3 Measurement-date discipline

ASC 718 measures equity-classified customer awards at **grant-date** fair value. Public filings do not disclose a grant-date valuation, and vesting schedules (8-K Exhibits E and F) are redacted. Main-paper magnitudes at March 2026 prices are therefore **post-grant sensitivities** that scale linearly with the measurement-date price—not a claim that grant-date fair value equals \$61.8 billion.

5.4 NVIDIA capacity backstop as a parallel ASC 606 question

NVIDIA’s commitment to purchase up to \$6.3 billion of unsold CoreWeave capacity through April 2032 (CoreWeave 10-K Note 15) is a service-form consideration-payable question under ASC 606-10-32-25, not an ASC 718 equity instrument. We flag it as an emerging parallel; full analysis is beyond this appendix.

6 Fire-sale recovery and short-life pricing

Main-paper §5.4 reports credit metrics and rental-rate evidence. This section extends the fire-sale and reduced-form pricing discussion.

6.1 Fire-sale counterargument and why GPUs differ from aircraft

Franks et al. (2022) decompose the raw 18% discount on aircraft sold by distressed airlines: under-maintenance explains roughly half; the residual 9–11% liquidity discount transfers value from seller to buyer rather than destroying it; Chapter 11 patient-sale provisions limit misallocation. That evidence warns against overstating welfare costs of fire sales. It does **not** rescue GPU collateral recovery:

1. **Welfare vs. recovery.** The lender recovers the realized price; quality and liquidity components both reduce recovery.
2. **Endogenous quality.** Covenant-stressed operators defer maintenance, cooling, and refurbishment exactly when the lender relies on collateral.
3. **Missing institutions.** No Section 1110 analog, no lessor class that redeploys hardware to final users, no standardized maintenance records, and generational displacement is a secular shock—collateral value need not mean-revert while a patient seller waits (Shleifer and Vishny 1992; Pulvino 1998).

Additional framework notes: `data/GPU_FIRE_SALE_RISK_FRAMEWORK.md`.

6.2 Pricing the short-life state

Bound the market-assigned probability of the short-life state with the reduced-form identity $\Delta s = p \times L$: the spread differential between GPU-backed and comparable credit compensates incremental expected loss on GPU collateral (Jarrow and Turnbull 1995; Duffie and Singleton 1999).

Two observed differentials:

Differential	Source	Implied short-life state price
~10 bp GPU-backed vs non-AI private credit	(Bank for International Settlements 2026)	At 40–60% loss severity: ~0.2–0.3% per year (<1.5% cumulative over five years)
400–600 bp concession avoided via Meta enhancement	(Bloomberg News 2026a)	~7–15% per year on the unenhanced alternative

Enhanced facilities price the short-life state near zero; the unenhanced market prices it an order of magnitude higher. Two contemporaneous prices on the same collateral class, differing only in which balance sheet absorbs the short-life state, identify that state price. Risk premia push risk-neutral figures above physical probabilities, so the 0.2–0.3% figure is an upper bound on physical probability under that spread. Duffie

and Lando (2001) predict wider spreads when investors observe asset value through imperfect accounting—the three measurement gaps in the main paper are that imperfection, yet average observed unenhanced-vs-enhanced wedges remain large.

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